# MILITARY STANDARD

MIL STD-1386-48
(NAVY)
11 NOVEMBER 1974

## LOADING OF HAZARDOUS MATERIALS AND ASSOCIATED ITEMS IN MILVAN CONTAINERS

### MINE UNDERWATER MK 55 AND MODS, CONFIGURATION "C"

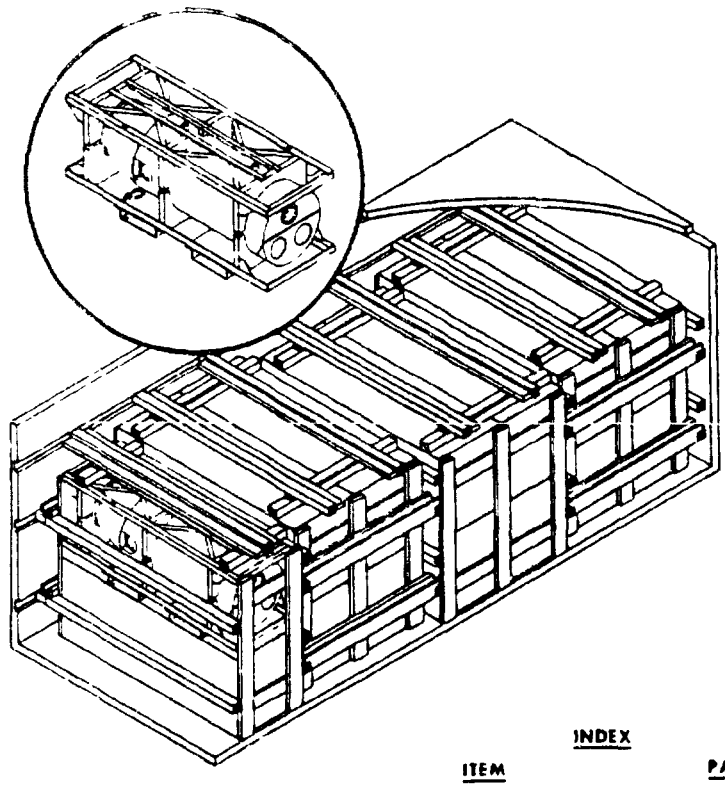
### IN CRATE MK 55 MOD 1

### FLEET ISSUE UNIT LOAD

**UNIT LOAD DATA**

| | |
|---|---|
| UNIT LOAD DRAWING | WR-54 7209 |
| DIMENSIONS | 84 L x 28 3/4 W x 32 1/4 H |
| CUBE | 45.0 CU FT |
| WEIGHT | SEE TABLE I |

HAZARD CLASSIFICATION
DOT EXPLOSIVES A
CC CLASS X-A



NOTES

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHE
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING TRUCKLOADING AND CARLOADING MILITARY STANDARDS REFER TO THE INDEX TO STANDARDS MIL-HDBK-236 (NAVY)

**INDEX**


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FSC 8140

**AUTHORIZED FOR TRAILER-ON-FLAT-CAR (TOFC), CONTAINER-ON-FLAT-CAR (COFC), CONTAINERSHIP AND HIGHWAY SHIPMENTS**

**APPROVED BY US COAST GUARD**

Acting CAPT USCG
11-7-74

**APPROVED BY BUREAU OF EXPLOSIVES**

for Director
24 Oct

USN Engr XMMH 11-5-74
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

SIGNATURE NAVSEA COM BY DIRECTION DATE

**ORIGINATOR** SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
NAD EARLE, NEW JERSEY



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# GENERAL NOTES

THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR LOADING THE MILVAN CONTAINER WITH MK 55 AND MODS CONFIGURATION C UNDERWATER MINES IN THE MK 55 MOD 1 CRATE THE MINES SHALL BE PALLETIZED IN ACCORDANCE WITH WB-54 7205

2 THE MILVAN SHALL BE EQUIPPED WITH A MECHANICAL LOAD-BRACING SYSTEM THAT MEETS THE REQUIREMENTS OF THE BUREAU OF EXPLOSIVES PAMPHLET 6C

3 IN ADDITION TO THE BELT RAILS REQUIRED BY PAMPHLET 6C THE MILVAN CONTAINER HAS BELT RAILS AT THE 72 AND 82 HEIGHTS FROM THE CONTAINER FLOOR

4 CONTAINERS ARE EQUIPPED WITH 25 CROSSMEMBER UNUSED CROSSMEMBERS MUST REMAIN WITH THE CONTAINER AND SHALL BE FASTENED INTO BELT RAILS FOR SHIPMENT

**CAUTION**

WHEN POSITIONING CROSSMEMBERS MAKE A VISUAL CHECK TO ENSURE THAT CROSSMEMBERS ARE LOCKED SECURELY IN PLACE

5 WHEN THE LONGITUDINAL SLACK BETWEEN A CROSSMEMBER AND THE UNIT LOAD IS GREATER THAN ONE INCH INSTALL SHIM PIECE 17 TO THE CROSSMEMBER TO FILL THE VOID TWIST TIE WITH THREE PIECES OF 16 GAUGE SOFT ANNEALED WIRE AS SHOWN IN DETAIL H THIS SHIM STRIP MAY BE ADDED OR OMITTED AS REQUIRED

6 THICKNESS OF HOLD DOWN ASSEMBLIES DETAIL F AND DETAIL G MAY BE VARIED TO ACCOMMODATE DIFFERENCES IN UNIT LOAD HEIGHTS BY VARYING THICKNESS OF SPACER AND TIE PIECES STRUCTURAL INTEGRITY OF HOLD DOWN ASSEMBLY SHALL BE MAINTAINED VOID BETWEEN HOLD DOWN CROSSMEMBER AND PIECE 11 OF DETAIL F AND PIECE 14 OF DETAIL G OF HOLD DOWN ASSEMBLIES SHALL NOT EXCEED ONE HALF INCH LOCATE SO THAT 4 x 4 S OF ASSEMBLY ARE OVER SADDLE OF CRATE

7 SWAY BRACES DETAIL A AND DETAIL B ARE TWIST TIED BOTH ENDS TO THE CROSSMEMBERS WITH 16 GAUGE TIE WIRE LENGTH OF STRUT PIECE 3 SHALL BE OF SUFFICIENT THICKNESS TO CAUSE THE SWAY BRACE TO BEAR AGAINST THE SIDE FRAME AND THE CONTAINER WALL

8 APPLICABLE MATERIAL SPECIFICATIONS
DUNNAGE LUMBER FED SPEC MM-L-751
NAILS - FED SPEC FF-N-105 TYPE II, STYLE 10 COMMON BRIGHT

9 AFTER BLOCKING AND BRACING HAS BEEN INSPECTED ATTACH SHIPPING DOCUMENTS INSIDE THE CONTAINER IN AN ACCESSIBLE AREA CLOSE AND SEAL CONTAINER DOORS AND ATTACH APPLICABLE PLACARDS TO THE OUTSIDE OF THE CONTAINER FOR THE APPROPRIATE MODE OF TRANSPORTATION AS PRESCRIBED IN OP 2165 (VOL )

# TOFC/COFC, CONTAINERSHIP AND HIGHWAY

THE CONTAINER LOAD CONSISTS OF 14 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS DOCUMENT

7 THE LOADED MILVAN CONTAINER SHALL BE ON A 20 FT MILVAN CHASSIS EQUIPPED WITH DOUBLE BOGIES WHEN BEING MOVED IN TOFC SERVICE

5 WHEN THE LOADED CONTAINER IS IN TOFC OR COFC SERVICE IT SHALL BE ON FLAT RAILCARS THAT COMPLY WITH BUREAU OF EXPLOSIVES PAMPHLET 6C

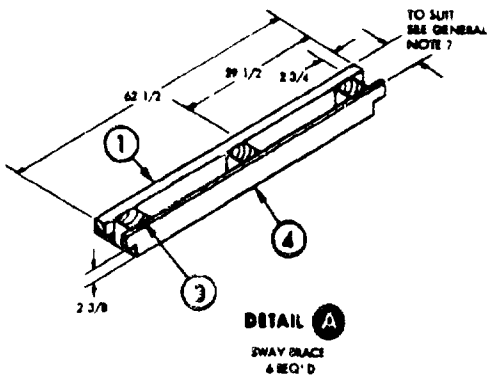
4 THE LOADED MILVAN MAY BE TRANSPORTED OVER PUBLIC HIGHWAYS ON A MILVAN CHASSIS EQUIPPED WITH DOUBLE BOGIES OR ON SUITABLE COMMERCIAL EQUIPMENT OF APPROPRIATE CAPACITY

**NOTE**

IN CERTAIN STATES THE GROSS VEHICLE WEIGHT (G V W ) IS LIMITED BY THE DISTANCE BETWEEN THE AXLES OF THE ASSEMBLED RIG CARE SHOULD BE TAKEN TO SEE THAT THE PROPOSED G V W DOES NOT EXCEED THESE LIMITATIONS

**TABLE I**

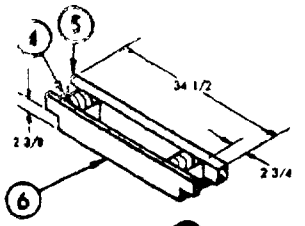
| MK 55 MINE UNDERWATER | UNIT LOAD WT (LBS) | CONTAINER LOAD WEIGHT (LBS) | CONTAINER GROSS WEIGHT (LBS) |
|---|---|---|---|
| MOD | 2305 | 32 270 | 38 644 |
| MOD 2 | 2375 | 33 306 | 39, 680 |
| MOD | 2389 | 33 446 | 39 820 |
| MOD 4 | 2318 | 32 452 | 38 826 |
| MOD | 2386 | 33, 404 | 39, 778 |
| MOD 6 | 2397 | 33 558 | 39 932 |
| MOD | 2168 | 30 352 | 36 726 |
| OD 8 | 2168 | 30 35 | 36 726 |




DETAIL A
SWAY BRACE 6 REQ'D

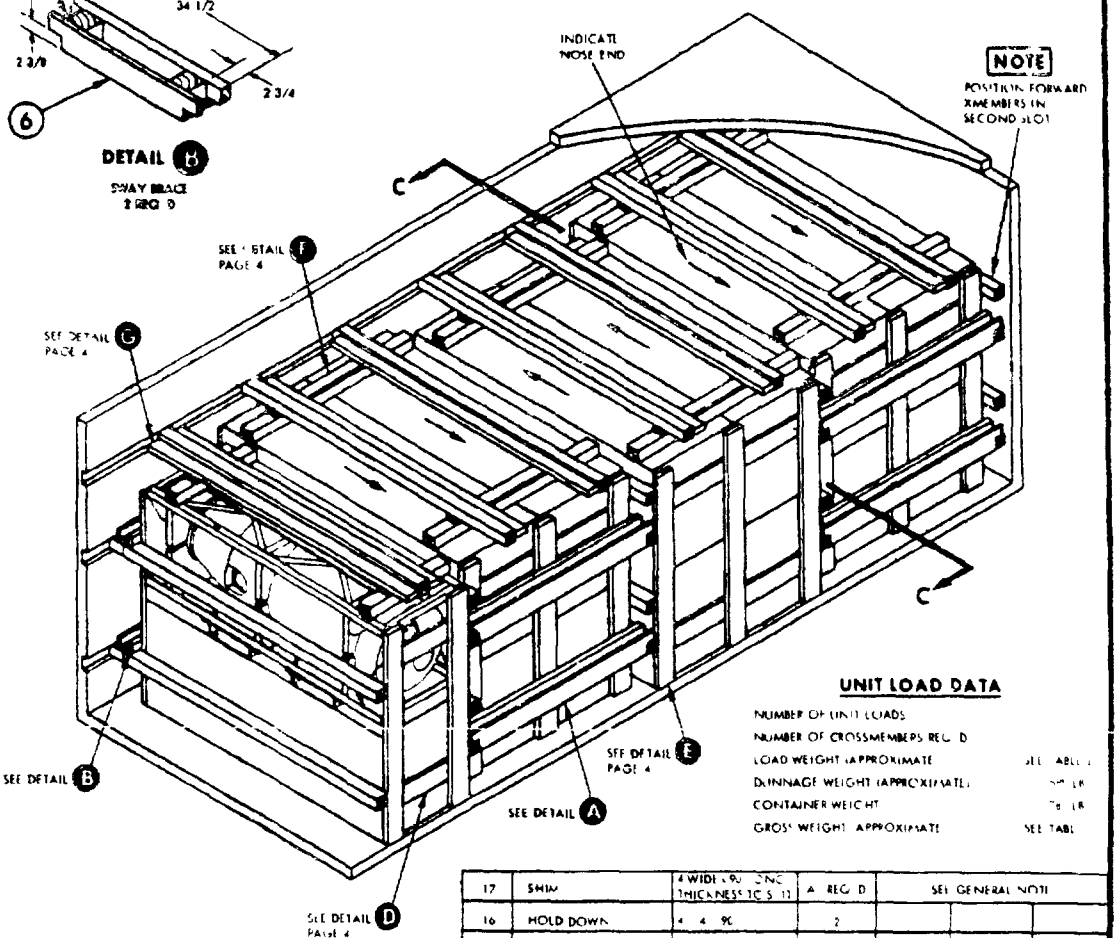


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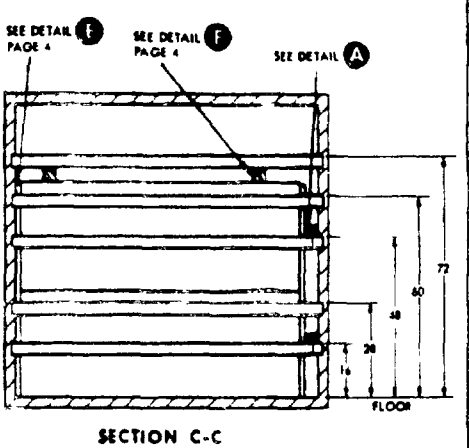

**DETAIL B**

SWAY BRACE
2 REQ'D




## UNIT LOAD DATA

| | |
|---|---|
| NUMBER OF UNIT LOADS | |
| NUMBER OF CROSSMEMBERS REQ'D | |
| LOAD WEIGHT (APPROXIMATE | SEE TABLE |
| DUNNAGE WEIGHT (APPROXIMATE) | [illegible] LB |
| CONTAINER WEIGHT | [illegible] LB |
| GROSS WEIGHT APPROXIMATE | SEE TABLE |



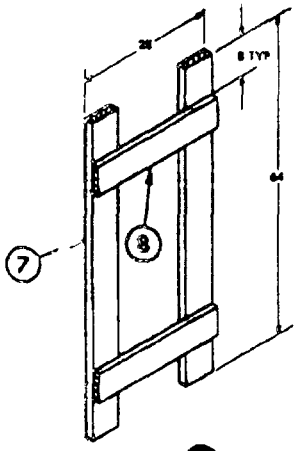

**SECTION C-C**

| PIECE NO | DESCRIPTION | SIZE | NO PCS REQ'D | NAIL TO | NAILS NUMBER | NAILS SIZE |
|---|---|---|---|---|---|---|
| 17 | SHIM | 4 WIDE x 90 LONG THICKNESS TO S[illegible] | A/R | | SEE GENERAL NOTE | |
| 16 | HOLD DOWN | 4 x 4 x 9[illegible] | 2 | | | |
| 15 | SPACER | 4 x 10 | 4 | 1 | JOINT | [illegible] |
| 14 | TIE PIECE | 4 x 4 x 90 | 2 | 14 | 3 JOINT | 3 |
| 13 | HOLD DOWN | 4 x 4 x 5 | 6 | | | |
| 12 | SPACER | [illegible] x 4 x 34 | 6 | 1 | 3 JOINT | 6d |
| 11 | TIE PIECE | [illegible] x 6 | 6 | 1 | JOINT | 16 |
| 10 | VERTICAL | 4 x 6 x 64 | 18 | | | |
| 9 | HORIZONTAL | [illegible] | 1 | | [illegible] IN | 8d CLINCHED |
| 8 | HORIZONTAL | [illegible] x 6 x 28 | 4 | 6 | [illegible] JOIN | 60 CLINCHED |
| 7 | VERTICAL | [illegible] x [illegible] x 64 | 4 | | | |
| 6 | STRINGER | 1 x 6 x 34 1/4 | 4 | | 2 JOIN | 16d |
| 5 | STRING[illegible] | [illegible] x 6 x 34 | 2 | | JOIN | 3 |
| 4 | STRUT | 4 x 4 TO S[illegible] | 4 | | | |
| | STRUT | 4 x [illegible] | 2 | | | |
| 2 | STRINGER | [illegible] x 6 x [illegible] | 4 | | [illegible] IN | |
| 1 | STRINGER | | 2 | | [illegible] | |

LIST OF MATERIAL AND NAILING DATA

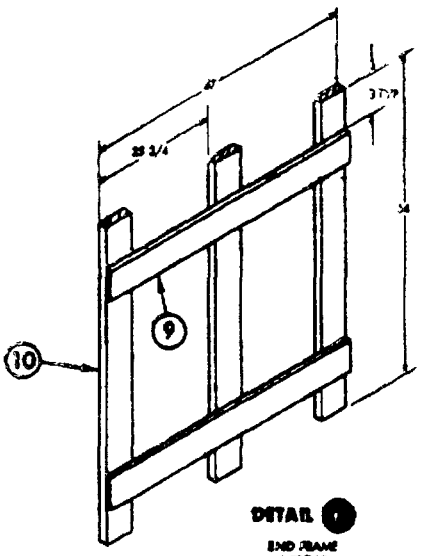


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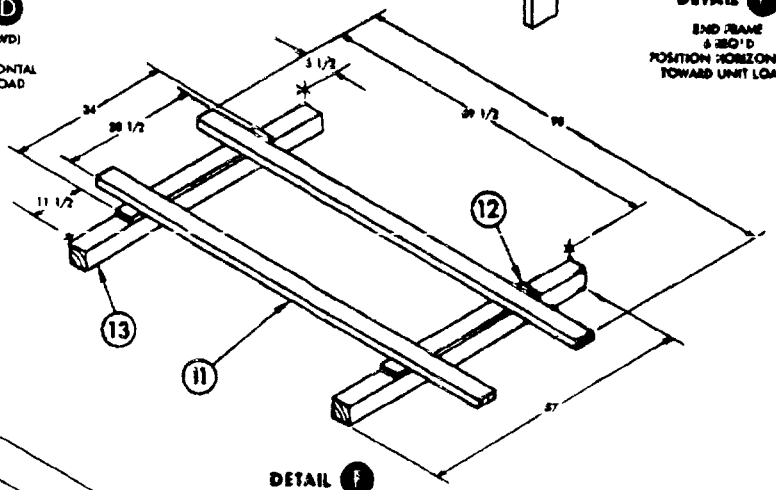

DETAIL D

END FRAME (FWD)
2 REQ'D
POSITION HORIZONTAL
TOWARD UNIT LOAD



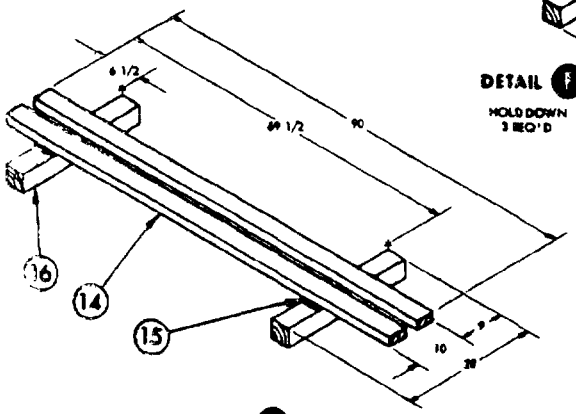

DETAIL E

END FRAME
6 REQ'D
POSITION HORIZONTAL
TOWARD UNIT LOAD



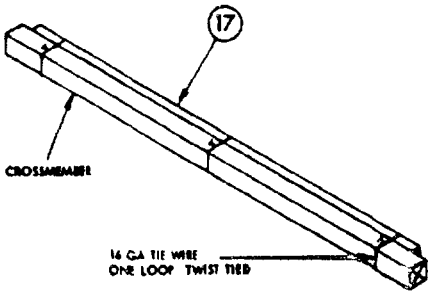

DETAIL F

HOLD DOWN
3 REQ'D




DETAIL G

HOLD DOWN
1 REQ'D




DETAIL H

FILL MATERIAL
SEE GENERAL NOTE 5

REVIEW ACTIVITY
NAVY OS

PREPARING ACTIVITY
NAVY OS
PROJECT NO 8140-N260

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FOLD

POSTAGE AND FEES PAID

OFFICIAL BUSINESS

Commanding Officer
Naval Weapons Station Earle
Colts Neck, New Jersey 07722

ATTN NWHL-803

FOLD

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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS This sheet is to be filled out by personnel either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side staple in corner and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION

MIL-STD-1386-48 (NAVY)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes" in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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